Stage 1 Hard Drill — With Solutions

Chapter 6.1 — Dynamic Programming on Random Variables

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the recursion is set up. Master rule: *start from the last decision and work backwards; the value of passing is the net you compare against.*

Q1 — An 8-Sided Die, Three Rolls

A fair 8-sided die may be rolled up to 3 times; after each roll you keep it or re-roll (the last roll is forced). Find the expected payoff and the accept rule at each roll.

Solution. $V_3 = \mathbb{E}[X] = \frac{9}{2}$. At roll 2 the net is 4.5: keep $\{5,6,7,8\}$ (sum 26); faces 1–4 become 4.5 (total 18):

$$V_2 = \frac{18+26}{8} = \frac{11}{2}$$

At roll 1 the net is 5.5: keep $\{6,7,8\}$ (sum 21); faces 1–5 become 5.5 (total 27.5):

$$V_1 = \frac{27.5+21}{8} = \boxed{\frac{97}{16} = 6.0625}$$

Rule: roll 1 keep ≥ 6; roll 2 keep ≥ 5; roll 3 forced.

⋆ *Intuition:* the staircase of nets $4.5 \to 5.5 \to 6.0625$ rises as more re-rolls remain, so your bar for "good enough" is highest at the start and drops each time the safety net shrinks.

Q2 — A Nonlinear Payoff

A fair d6 may be rolled up to 3 times, but the payoff is the *square* of the kept face. Find the expected payoff and compare with $(14/3)^2$.

Solution. Run the recursion on the *payoff* X^2, not on X. $V_3 = \mathbb{E}[X^2] = \frac{91}{6} \approx 15.17$.

Roll 2: keep iff $X^2 > 15.17 \iff X \geq 4$ (payoffs $16, 25, 36$, sum 77); the other three faces become $\frac{91}{6}$:

$$V_2 = \frac{3\cdot\frac{91}{6}+77}{6} = \frac{245}{12} \approx 20.42$$

Roll 1: keep iff $X^2 > 20.42 \iff X \geq 5$ (payoffs $25, 36$, sum 61):

$$V_1 = \frac{4\cdot\frac{245}{12}+61}{6} = \boxed{\frac{214}{9} \approx 23.78}$$

Compare: $\left(\frac{14}{3}\right)^2 = 21.78$. The gap 2.0 is exactly $\mathrm{Var}(X^\star)$ for the final kept face $X^\star$.

▶ *Speed trick:* never apply the function to the *value*: $\mathbb{E}[f(X^\star)] \neq f(\mathbb{E}[X^\star])$. Convex payoffs reward the optimiser for the spread of the outcome as well as its mean.

Q3 — Maximising a Win Probability, Not a Mean

You roll a d6 with one optional re-roll; a friend then rolls once and the higher number wins (a tie is a coin flip). (a) Maximise $\mathbb{P}(\text{win})$. (b) Redo it if the friend rolls *two* dice and keeps the larger.

Solution. Let $g(x) = \mathbb{P}(\text{win} \mid \text{your final} = x)$ and run the DP on g. (a) $g(x) = \frac{x-1}{6} + \frac{1}{2}\cdot\frac{1}{6} = \frac{2x-1}{12}$, which is linear in x, so the same rule as in the plain game applies (keep ≥ 4):

$$\mathbb{P}(\text{win}) = \mathbb{E}[g] = \frac{2(4.25)-1}{12} = \boxed{\frac{5}{8}}$$

(b) For the max of two dice, $\mathbb{P}(M < x) = \frac{(x-1)^2}{36}$ and $\mathbb{P}(M = x) = \frac{2x-1}{36}$:

$$g(x) = \frac{2x^2-2x+1}{72} \in \left\{\tfrac{1}{72}, \tfrac{5}{72}, \tfrac{13}{72}, \tfrac{25}{72}, \tfrac{41}{72}, \tfrac{61}{72}\right\}$$

Final roll: $\mathbb{E}[g] = \frac{146}{432} = \frac{73}{216}$. Keep iff $g(x) > \frac{73}{216}$, i.e. $x \geq 4$ (as $\frac{25}{72} > \frac{73}{216}$):

$$V_1 = \frac{3\cdot\frac{73}{216}+\frac{25+41+61}{72}}{6} = \boxed{\frac{25}{54} \approx 0.463}$$

⋆ *Intuition:* whatever you are optimising, put *that* quantity at the leaves of the tree. Only when it is linear in the face do "maximise the mean" and "maximise the win probability" coincide.

Q4 — Buying at the Lowest Quote (Minimisation)

You must buy and receive up to 3 price quotes U_1, U_2, U_3 iid Uniform$(0,1)$, one at a time; you may take the current quote or pass (the last is forced). Find the expected price paid and the thresholds.

Solution. Minimise: the net is now an upper bar t, and quotes above it are replaced by t:

$$\mathbb{E}[\min(U,t)] = t - \int_0^t \mathbb{P}(U \leq s)\,ds = t - \frac{t^2}{2}$$

$$V_3 = \tfrac{1}{2}, \qquad V_2 = \tfrac{1}{2} - \tfrac{1}{8} = \tfrac{3}{8}$$

$$V_1 = \tfrac{3}{8} - \tfrac{(3/8)^2}{2} = \tfrac{48-9}{128} = \boxed{\tfrac{39}{128} \approx 0.305}$$

Rule: accept quote 1 iff $< \frac{3}{8}$; quote 2 iff $< \frac{1}{2}$.

Mirror check: with $Y = 1-U$ the sell-side game has $V_1 = \frac{89}{128}$, and $1 - \frac{89}{128} = \frac{39}{128}$ ✓.

▶ *Speed trick:* a buy-side problem is the sell-side problem reflected: solve one, subtract from the upper bound for the other.

Q5 — Normally Distributed Offers

Offers are iid $\mathcal{N}(0,1)$, at most 3 draws, accept or pass. Find V_1 and the thresholds.

Solution. For a net t, with φ, Φ the standard normal density and CDF:

$$\mathbb{E}[\max(Z,t)] = t\,\Phi(t) + \int_t^\infty z\varphi(z)\,dz = t\,\Phi(t) + \varphi(t)$$

using $\int_t^\infty z\varphi(z)\,dz = \varphi(t)$. Then

$$V_3 = 0, \qquad V_2 = \varphi(0) = \tfrac{1}{\sqrt{2\pi}} \approx 0.3989$$

$$V_1 = 0.3989\cdot\underbrace{\Phi(0.3989)}_{0.6550} + \underbrace{\varphi(0.3989)}_{0.3684} \approx \boxed{0.630}$$

Rule: draw 1 accept iff $Z > 0.399$; draw 2 accept iff $Z > 0$.

⋆ *Intuition:* a clairvoyant who sees all three draws and takes the best earns $\mathbb{E}[\max] \approx 0.846$; the stopper, who must commit as each draw arrives, earns 0.63.

Q6 — How Many d12 Rolls Equal One d20 Roll?

One roll of a d20 is worth 10.5. A fair d12 can be re-rolled any number of times up to a cap of k rolls. What is the smallest k whose optimal value exceeds 10.5?

Solution. With net t and $m = \lfloor t \rfloor$ faces replaced by t (the faces sum to 78):

$$V_{\text{new}} = \frac{m\,t + \left[78 - \frac{m(m+1)}{2}\right]}{12}$$

k	1	2	3	4	5	6	7	8
V_1	6.5	8	8.83	9.39	9.79	10.09	10.33	10.52

For example, $k=4$: $t = \frac{53}{6}$, $m=8$: $\frac{8\cdot 53/6+42}{12} = \frac{169}{18}$. At $k=7$ the value is $10.33 < 10.5$ and at $k=8$ it is $\frac{1347}{128} \approx 10.52 > 10.5$:

$$\boxed{k = 8}$$

▶ *Speed trick:* each re-roll buys less than the one before (Example 3's diminishing returns). Eight d12 rolls with optimal stopping barely match a single d20 roll — *a bigger die beats more tries.*

Q7 — The Casino Might Close

Roll a d6 repeatedly. Each time you pass, there is a 25% chance the casino closes (payoff 0) and otherwise you roll again. Find the value, the optimal rule and $\mathbb{P}(\text{end with } 0)$.

Solution. Passing is worth δV with $\delta = \frac{3}{4}$, so $V = \mathbb{E}[\max(X, \frac{3}{4}V)]$.

Guess that faces $1, 2$ are rejected ($\frac{3}{4}V \in [2, 3]$):

$$V = \frac{2 \cdot \frac{3}{4}V + 3 + 4 + 5 + 6}{6} \;\Rightarrow\; 6V = 1.5V + 18 \;\Rightarrow\; V = \boxed{4}$$

Check: $\frac{3}{4}V = 3$ lies in $[2, 3]$ ✓, so keep $X \geq 3$ (a 3 is a tie). $\mathbb{P}(\text{keep}) = \frac{2}{3}$ per roll; reject $\frac{1}{3}$, then survive with $\frac{3}{4}$:

$$\mathbb{P}(\text{paid}) = \frac{2/3}{1 - \frac{1}{3} \cdot \frac{3}{4}} = \frac{8}{9} \;\Rightarrow\; \mathbb{P}(\text{end with } 0) = \boxed{\tfrac{1}{9}}$$

Check: $\frac{8}{9} \cdot \mathbb{E}[X \mid X \geq 3] = \frac{8}{9} \cdot 4.5 = 4$ ✓.
⋆ *Intuition:* "risk of the game ending" is just a discount factor δ. Same value 4, different risk: breaking the tie the other way (reject a 3) also gives $V = 4$ but $\mathbb{P}(0) = \frac{1}{5}$.

Q8 — Roll or Bank (a Bust Game)

Roll a fair d6 repeatedly. Faces 2–6 are added to your bank. A face 1 *busts*: you lose the bank, get 0, and the game ends. You may stop at any time and keep the bank. (a) What is the optimal stopping rule? (b) Find the values of banks 19 and 18. (c) What is the expected payoff starting from 0?
Solution. (a) From bank b, one more roll gives bank $b + f$ w.p. $\frac{1}{6}$ each for $f = 2, \ldots, 6$ and 0 w.p. $\frac{1}{6}$:

$$\mathbb{E}[\text{new bank}] = \frac{5b + 20}{6}, \qquad \text{gain} = \frac{5b + 20}{6} - b = \frac{20 - b}{6}$$

The gain is positive iff $b < 20$ and strictly shrinks as b grows, so once stopping is better it is better for every larger bank. That monotonicity makes the *one-step look-ahead* rule optimal: **roll while $b < 20$, stop at $b \geq 20$.**
(b) $b = 19$: every non-bust roll lands ≥ 21, so stop there:

$$V(19) = \tfrac{1}{6}(21 + 22 + 23 + 24 + 25) = \boxed{\tfrac{115}{6} \approx 19.17}$$

$b = 18$: rolls land on $20, \ldots, 24$:

$$V(18) = \tfrac{1}{6}(20 + 21 + 22 + 23 + 24) = \boxed{\tfrac{55}{3} \approx 18.33}$$

(c) Backward recursion $V(b) = \frac{1}{6}\sum_{f=2}^{6} V(b+f)$ for $b < 20$, $V(b) = b$ for $b \geq 20$, down to $b = 0$:

$$V(0) \approx \boxed{8.14}$$

(simulation: 8.14). You reach ≥ 20 only about 37.5% of the time, averaging ≈ 21.7 when you do.
▶ *Speed trick:* bust games are solved by comparing *expected gain to expected loss*: $\frac{5}{6} \cdot \mathbb{E}[f \mid f \geq 2] = \frac{20}{6}$ against $\frac{1}{6}b$ gives the target $b^\star = 20$ at once.

Q9 — Gambler vs Prophet

(a) With a d6 and 2 rolls, compare the optimal stopper's value with that of a *prophet* who sees both rolls and takes the larger. (b) Repeat with 3 iid Uniform$(0, 1)$ offers.
Solution. (a) Stopper: $\frac{17}{4} = 4.25$ (Example 1). Prophet: $\mathbb{P}(\max \geq k) = 1 - \frac{(k-1)^2}{36}$, so

$$\mathbb{E}[\max] = \sum_{k=1}^{6} \left(1 - \tfrac{(k-1)^2}{36}\right) = 6 - \tfrac{55}{36} = \tfrac{161}{36} \approx 4.472$$

$$\text{ratio} = \frac{17/4}{161/36} = \frac{153}{161} \approx \boxed{0.950}$$

The price of not seeing the future is only $\frac{161-153}{36} = \frac{2}{9} \approx 0.22$.
(b) Stopper: $\frac{89}{128}$. Prophet: $\mathbb{E}[\max] = \frac{3}{4}$:

$$\text{ratio} = \frac{89/128}{3/4} = \frac{89}{96} \approx \boxed{0.927}$$

⋆ *Intuition:* the worst-case guarantee for iid offers (the *prophet inequality*, Hill–Kertz) is about 0.745 of the prophet's value. These ratios sit well above it, because the offers here are mild and a good threshold gets you almost all of the maximum.

Q10 — Capstone: When Is Unlimited Patience Worth It?

Game A: three *free* rolls of a d6 (value $\frac{14}{3}$). Game B: unlimited rolls, but *every* roll (including the first) costs c, and you keep the last roll. At what cost $c^\star$ are the two games equally valuable? Check your answer by direct calculation.
Solution. In Game B the value C satisfies $C = -c + \mathbb{E}[\max(X, C)]$, i.e.

$$\mathbb{E}\big[(X - C)^+\big] = c$$

Set $C = \frac{14}{3} \in [4, 5)$: only 5 and 6 exceed it, so

$$c^\star = \frac{(5 - \frac{14}{3}) + (6 - \frac{14}{3})}{6} = \frac{\frac{1}{3} + \frac{4}{3}}{6} = \boxed{\tfrac{5}{18} \approx 0.278}$$

Direct check: at $c = \frac{5}{18}$ you stop on $\{5, 6\}$, so $\mathbb{E}[\#\text{rolls}] = \frac{1}{1/3} = 3$ and

$$\text{value} = \mathbb{E}[X \mid X \geq 5] - 3c = 5.5 - \tfrac{5}{6} = \tfrac{14}{3} \;\checkmark$$

C decreases as c rises, so Game B beats Game A iff $c < \frac{5}{18}$.
▶ *Full-circle intuition:* the finite horizon (a cap on tries) and the per-try price (a cost per try) are two ways of rationing patience, and the same backward-induction idea converts one into the other. Here a 0.28 fee per roll is exactly the exchange rate for "unlimited tries" ↔ "three free tries".

Drill order for review: redo Q2 and Q3 closed-book (run the DP on the *payoff*, not the face), then Q7 and Q8 (a risk of ending acts like a discount; a bust game is solved by gain versus loss), and finally Q10 (explain why a per-roll fee and a cap on rolls are interchangeable). Together they test whether the backward recursion, the cost equation $\mathbb{E}[(X - C)^+] = c$ and one-step look-ahead work as one connected skill.